Mohr-Coulomb Shear Strength & Shallow Foundation Design

Direct shear and triaxial shear test interpretations, Terzaghi's bearing capacity equations, consolidation settlement, and footing design.

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1. Mohr-Coulomb Failure Criterion

Shear strength tau_f = c' + sigma' * tan(phi'), balancing effective cohesion c' and internal friction angle phi' derived from triaxial tests.

2. Terzaghi's Bearing Capacity Equations

For shallow footings, ultimate bearing capacity qu = c*Nc + gamma*Df*Nq + 0.5*gamma*B*N_gamma with safety factor 2.5 to 3.0.